



B01 PROTECTED PRACTICE REVIEW

Open a lesson review only after that lesson has a saved genuine attempt.

Protection rule

This review repairs reasoning. It is not the independent attempt. If you opened a review early, mark the attempt contaminated and complete a fresh changed task before mastery.

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B01-L01 Review - Grain and cardinality contracts for engineering SQL

Unlock condition

Open only after P-B01-L01 has a saved genuine attempt, validation notes and an explain-back attempt.

What strong work demonstrates

Strong work declares every grain before the join, predicts one-to-many fan-out, aggregates to the desired grain before attaching non-additive/order-level fields, and uses more than one control.

Repair the failed skill

If order_id repeats, first ask whether the current output is item-grain. If shipping_fee is summed after an item/promotion join, compare against an order-only shipping control. Never use DISTINCT until you can explain exactly which business rows are duplicates.

Do not copy a final answer

This review intentionally does not provide a paste-ready solution. Compare your reasoning to the contracts above, repair only the weak skill, then close the review.

Changed retry

Fresh transfer

Changed retry: Build one row per customer-month from completed orders and item rows. Prove that a multi-row promotions source cannot multiply customer-month revenue.

Ready to mark Mastered when

The changed retry is correct, independently validated, and explainable without this review open. Save the earlier failed attempt; history is append-only.

B01-L02 Review - Staged transformations with CTEs and subqueries

Unlock condition

Open only after P-B01-L02 has a saved genuine attempt, validation notes and an explain-back attempt.

What strong work demonstrates

Strong work uses stage names that describe purpose, not cte1/cte2. Every stage has one declared grain. At least one intermediate result is executed alone and validated.

Repair the failed skill

If the final total is wrong, do not rewrite the whole query. Run stages in order and locate the first point where row/key/financial invariants change unexpectedly.

Do not copy a final answer

This review intentionally does not provide a paste-ready solution. Compare your reasoning to the contracts above, repair only the weak skill, then close the review.

Changed retry

Fresh transfer

Changed retry: For a different date window, build item_base -> order_level -> customer_summary and identify above-average customers. Validate order_level independently.

Ready to mark Mastered when

The changed retry is correct, independently validated, and explainable without this review open. Save the earlier failed attempt; history is append-only.

B01-L03 Review - Window patterns for ranking, latest-row and sequence logic

Unlock condition

Open only after P-B01-L03 has a saved genuine attempt, validation notes and an explain-back attempt.

What strong work demonstrates

Strong work separates result ordering from window ordering, uses a complete tie-break when row selection/ranking must be deterministic, and states whether LAG means previous observed row or previous calendar period.

Repair the failed skill

If rank changes between runs on ties, add a stable ordering key. If a running total behaves strangely, inspect the frame. If LAG skips a month, create a calendar series later only when the business requirement truly needs missing months represented.

Do not copy a final answer

This review intentionally does not provide a paste-ready solution. Compare your reasoning to the contracts above, repair only the weak skill, then close the review.

Changed retry

Fresh transfer

Changed retry: Rank products within region by completed revenue with a stable tie-break and compare each month to the previous observed month.

Ready to mark Mastered when

The changed retry is correct, independently validated, and explainable without this review open. Save the earlier failed attempt; history is append-only.

B01-L04 Review - Conditional and date transformations with explicit business rules

Unlock condition

Open only after P-B01-L04 has a saved genuine attempt, validation notes and an explain-back attempt.

What strong work demonstrates

Strong work writes business definitions before SQL: numerator, denominator, included statuses, start/end timestamps and output time grain.

Repair the failed skill

If the rate seems wrong, calculate one group manually. For time windows, test a row exactly on each boundary. Prefer adjacent half-open windows when repeated processing must avoid overlap.

Do not copy a final answer

This review intentionally does not provide a paste-ready solution. Compare your reasoning to the contracts above, repair only the weak skill, then close the review.

Changed retry

Fresh transfer

Changed retry: Compute a different status rate and count events in a new [start,end) interval. Prove boundary behavior with min/max and one exact endpoint test.

Ready to mark Mastered when

The changed retry is correct, independently validated, and explainable without this review open. Save the earlier failed attempt; history is append-only.

B01-L05 Review - Data-quality investigation queries

Unlock condition

Open only after P-B01-L05 has a saved genuine attempt, validation notes and an explain-back attempt.

What strong work demonstrates

Strong work makes each quality query correspond to a stated rule and records impact/severity. It keeps evidence instead of silently cleaning it away.

Repair the failed skill

For duplicates, distinguish duplicate business event from legitimate multiple child rows. For orphans, consider whether the parent can arrive late. BLOCK/WARN depends on the trust contract, not emotion.

Do not copy a final answer

This review intentionally does not provide a paste-ready solution. Compare your reasoning to the contracts above, repair only the weak skill, then close the review.

Changed retry

Fresh transfer

Changed retry: Define a new business-key duplicate rule on a changed combination and one new range rule. Prove a known-good row is not falsely flagged.

Ready to mark Mastered when

The changed retry is correct, independently validated, and explainable without this review open. Save the earlier failed attempt; history is append-only.

B01-L06 Review - Reconciliation and validation SQL

Unlock condition

Open only after P-B01-L06 has a saved genuine attempt, validation notes and an explain-back attempt.

What strong work demonstrates

Strong validation can disagree with the transformation because it follows a different path or checks a different invariant. It identifies the first failing stage.

Repair the failed skill

If a control always passes even when you restore bad logic, it is not protecting you. Design the control from the failure mode: missing rows, extra rows, multiplied money, wrong population or wrong time boundary.

Do not copy a final answer

This review intentionally does not provide a paste-ready solution. Compare your reasoning to the contracts above, repair only the weak skill, then close the review.

Changed retry

Fresh transfer

Changed retry: Create a new fan-out defect using a different one-to-many source. Write an independent control first, prove FAIL, then repair and prove PASS.

Ready to mark Mastered when

The changed retry is correct, independently validated, and explainable without this review open. Save the earlier failed attempt; history is append-only.

B01-L07 Review - EXPLAIN, scans and index awareness

Unlock condition

Open only after P-B01-L07 has a saved genuine attempt, validation notes and an explain-back attempt.

What strong work demonstrates

Strong work treats EXPLAIN as evidence, not a scorecard. It identifies scan/filter/estimate facts, proposes one index from the query pattern, and keeps correctness separate from speed.

Repair the failed skill

A Seq Scan can be rational for a tiny table. Cost is not milliseconds. An index that exists may not be chosen. Do not force the planner or claim improvement without measurement.

Do not copy a final answer

This review intentionally does not provide a paste-ready solution. Compare your reasoning to the contracts above, repair only the weak skill, then close the review.

Changed retry

Fresh transfer

Changed retry: EXPLAIN a different selective predicate/order requirement. Decide whether an index proposal is justified and write both the evidence and uncertainty.

Ready to mark Mastered when

The changed retry is correct, independently validated, and explainable without this review open. Save the earlier failed attempt; history is append-only.

B01-L08 Review - Maintainable deterministic SQL handoff

Unlock condition

Open only after P-B01-L08 has a saved genuine attempt, validation notes and an explain-back attempt.

What strong work demonstrates

Strong handoff begins with purpose and contracts. It uses explicit columns, meaningful stages, deterministic order when order matters, separate validation, known limitations and an owner/next action.

Repair the failed skill

If another person needs to ask what one row means, what dates are included or why a join is safe, the handoff is incomplete. Comments should explain business meaning and risk, not narrate obvious syntax.

Do not copy a final answer

This review intentionally does not provide a paste-ready solution. Compare your reasoning to the contracts above, repair only the weak skill, then close the review.

Changed retry

Fresh transfer

Changed retry: Produce a segment-month handoff with a different final grain and quality risk. The new reader must be able to validate it without your verbal help.

Ready to mark Mastered when

The changed retry is correct, independently validated, and explainable without this review open. Save the earlier failed attempt; history is append-only.